Akbar Abilov

Master of Science Student in Urban Data Science, New York University | F-1 OPT, STEM OPT eligible
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Focus Areas

- LiDAR point clouds for road and pavement condition
- Computer vision and geospatial machine learning
- Python and PostgreSQL backends and data pipelines
- Data systems for public infrastructure

Research Experience

Student Researcher, NYU Vertically Integrated Projects course, Prof. Laefer – Brooklyn, NY Jan 2026 – Present

Project: Aerial LiDAR and hyperspectral imaging for urban roadway condition assessment (Jan 2026 – May 2026)

- Detected 86 of 105 tree canopies (82%) from aerial LiDAR intensity alone by separating ground points with a cloth simulation filter in CloudCompare and filtering shadows in Python, in a 3-person team
- Built 2 clustering pipelines for pavement repair detection, K-means with principal component analysis and point-cloud descriptors with DBSCAN, and validated both against NYC excavation permits
- Extracted lane markings and matched LiDAR-detected manholes with hyperspectral material classes, and found that robust intensity normalization made pavement variation more visible

Project: LiDAR mapping of surfaces people touch after leaving NYC hospitals (Sept 2026 – Present)

- Built a first 3D touch-point heatmap on a LiDAR point cloud of NYU Langone Brooklyn by matching recorded touches to NYC bus stop open data in Python at a realistic hand height, in a 3-person team

Student Researcher, NYU Guided Study course, Prof. Ergan – Brooklyn, NY Sept 2025 – Dec 2025

- Annotated frame by frame how 40 participants reacted to 6 virtual reality work zone scenarios (2–3 minutes each) modeled on past crash conditions, and cleaned the participant dataset, in a 2-person team

Entrepreneurship

Co-founder, Arxitech Solutions Dec 2025 – June 2026

Revit and AutoCAD automation for engineering firms; NYU Startup Bootcamp (Dec 2025) and Sprint (May 2026)

- Ran 19 customer discovery interviews with 18 engineers and architects, then pivoted the target customer from BIM architects to VDC engineers and BIM specialists, who showed the strongest pain
- Found that price-sensitive BIM managers approve every new Revit add-in, in a 2-person founding team

Projects

Illegal Dumping Detection, NYU Capstone Sept 2026 – Present

AI detection of illegal dumping from street sweeper camera footage, sponsored by the City of Paterson, NJ

- Mapped 1,012 high foot-traffic places in Paterson from NJ Transit, USDA, and OpenStreetMap data in Python and defined 16 survey zones, 2 of them must-survey, to guide the city's litter survey, in a 3-person team
- Profiled all 40 census tracts on 14 Census indicators with reliability flags, and grouped them by income and population mix so the survey covers every type of neighborhood
- Led all communication with the city sponsor and course team as the team's communications lead, setting agendas and running sponsor meetings

Pull, Congressional Bill Tracker github.com/NYU-Congressmen/congress-tracker

- Recovered 90 of 19,832 bills silently dropped on every sync by tracing unstable offset paging over 2,023 tied sort dates and adding a count-checked re-read with deduplication in Python
- Reduced bill summary sync from about 19,700 API requests to 26 by switching to a date-ranged bulk endpoint, loading all 6,446 summaries within a 5,000-request hourly limit
- Designed 14 PostgreSQL tables on Supabase across 3 Alembic migrations with CHECK, UNIQUE, and foreign key constraints, and built FastAPI endpoints covered by 83 pytest tests in GitHub Actions, in a 4-person team

Satellite Land-Use Classification github.com/ekberby/Deep-Learning-Land-Use

- Reached 92.6% test accuracy on 27,000 EuroSAT satellite images across 10 land-use classes, with 97–99% precision on forest and water, by fine-tuning a pretrained ResNet18 in PyTorch

NYC Bus Speed Prediction

- Predicted bus speed at R^2 0.704 (vs 0.435 baseline) and severe delays at ROC-AUC 0.900 by merging 5 transit datasets into a 6.14M-row pandas table and training scikit-learn random forests in a 4-person team

NYC EV Charging Access Analysis

- Found that only 59 of 2,327 NYC census tracts (2.5%) have a public Level 2 charger by building a reproducible Python pipeline in GeoPandas and OSMnx that joins 6 public datasets, in a 2-person team

AskNYC Zoning Assistant github.com/ekberby/AskNYC

- Built a retrieval-augmented generation (RAG) assistant in Python that cites a zoning section for every claim, packaged in Docker with a locally hosted Llama 3.1 model

Work Experience

Data Platform Developer, NYU Marron Institute of Urban Management – New York, NY Oct 2026 – Present

- Architected a 10-module system and relational data model that unifies case management, project management, and monitoring and evaluation for field programs handling sensitive personal data, replacing Excel-based tracking
- Designed offline data sync and role-based access control with consent checks, so field staff can record data without connectivity while personal records stay protected
- Translated 47 stakeholder requirements and 6 use cases into a design specification, tracing 90 design changes to their source

BIM Automation Intern, PortBIM May 2025 – Aug 2025

- Reduced manual engineering workflow time by 60–70% by building 5+ automation tools for Autodesk Revit and AutoCAD in C# and Python
- Improved automation speed 2–3x by refactoring Autodesk Dynamo visual scripts into compiled C# code
- Saved engineers 10+ hours per week by building a centralized update and configuration system that deployed new tool versions to every workstation at once

Education

New York University – Master of Science in Urban Data Science Expected May 2027
Relevant courses: Code-Based GIS and GeoAI, Machine Learning for Cities, Applied Data Science, Civic Analytics

University of Essex – Bachelor of Science in Computer Science July 2025

Technical Skills

Languages: Python, SQL, C#, Java
Machine Learning and Geospatial: PyTorch, scikit-learn, pandas, GeoPandas, OSMnx, CloudCompare
Backend and Tools: FastAPI, PostgreSQL, Supabase, Alembic, pytest, Docker, GitHub Actions, Claude Code

Certificates

Data Governance, Ethics, and Privacy Micro-badge, NYU Center for Urban Science + Progress Jan 2026